

CMP6200

Individual Honors Project

AI-Powered Virtual Tour Guide with Augmented Reality for Cultural Landmarks

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0.1 Introduction and Context

Cultural tourism is useful in preserving heritage, enhancing cross-cultural understanding, and strengthening local economies. Kathmandu Valley is home to numerous UNESCO World Heritage Sites and it is a popular destination by foreigners each year. Nevertheless, there is a lot of self-exploration of these sites by tourists, and in this case, they do not necessarily understand the historical, cultural, or architectural value (OECD, 2020).

New forms of digital tourism, like the straightforward mobile applications and audio tour guides tend to give general information and limited engagement as well as customization to the needs of the visitor (Bakas et al., 2019). Although recent developments in AI, AR, CV, and NLP have the ability to enhance experiences of being on-site in tourism, using these technologies completely is not sometimes done as a system but rather as individual elements (Chung et al., 2021).

The proposed project is the Smart Virtual Tour Guide mobile application that is powered by AI and which is targeted primarily at foreign visitors to heritage sites in Kathmandu Valley. The system will be targeted at the independent traveler without professional guides, integrating CV-based landmark detection, markerless AR overlay, an NLP-based conversational assistant, and customized recommendation algorithm to give the user an interactive and context-aware cultural tourism experience.

0.2 Problem Statement

The following key problems motivate this project:

- **Limited access to interactive and multilingual cultural guidance:**

Kathmandu Valley lacks good tour guides as offered to foreign tourists. The only major sites to get guide tours are not everywhere on a regular basis and are mainly available only during busy periods or large places of interest. Most of the guides either speak little English or not at all, hence the non-English speaking people receive minimal cultural guides. Travelers will then apply ripped sources such as signs or online searches, which are not interactive and that are incomplete. This renders the visits less interesting, less informative, and damages the tourism experience as a whole. We require a tool that is scalable and provides appropriate cultural explanations at one minute, in numerous languages and responding to various curiosity levels.

- **Limitations of existing digital and AR-based tourism tools:**

The help of digital and AR tourism tools has emerged within recent years, yet the majority of them remain static and actually provide information. They have set

text, images or labels on the sites and do not worry about the interaction people want or the interests that people have or the way people have certain questions. Tourists are not able to ask follow-up questions, probe into history and seek explanations to suit their tastes. A great number of AR applications also do not require smart recognition, meaning that they will not react to what they see automatically, but rather require people to select content. This renders the learning process passive rather than active and reduces the capability of giving vivid cultural experiences of AR. This has seen existing tools fail to offer indeed interactive and responsive heritage guidance.

- **Fragmented integration of AI technologies in tourism applications:**

Regardless of the developments in AI such as computer vision, natural language processing, and AR, most tourism applications currently combine them into individual uses, rather than combined into one. The model of perception of a landmark recognition does not necessarily imply the use of AR visualization, and a chatbot does not necessarily interact with the space a tourist is in. This fragmented arrangement presents the customers with a fractured experience where recognition, display and interaction do not reinforce. Consequently, tourists do not enjoy the flow of information that helps to relate what they observe, what they request, and what they get to know. The absence of a holistic AI system restricts the usefulness of such techs in the actual heritage context. It is quite important to address this fragmentation to create a smart tourism application that is sensitive, interactive, and provides immersive cultural narratives in real-time

0.3 Review of Existing Knowledge

This section reviews prior research related to the application of Augmented Reality, Computer Vision, Natural Language Processing, and recommendation systems in cultural tourism. The objective is to identify existing approaches, highlight their contributions, and critically examine the limitations that motivate the proposed project.

0.4 Augmented Reality in Cultural Tourism

The application of Augmented Reality (AR) in the cultural tourism sector is trending due to the fact that it can enable visitors to be more engaged as digital information is placed on top of actual heritage sites. Azuma had established the initial concepts of AR in 1997. According to him, AR ought to be characterized by three primary characteristics namely: real time, placing the virtual objects where they belong in the three dimensional space

and also integrating the real and the virtual objects into each other. These concepts continue to be employed in the construction of current AR systems.

With cultural tourism, AR applications tend to display information such as historical information, dates, notes in the building or other visuals right when standing in a particular area. Studies indicate that AR guides have the ability to appreciate the experience of the visitors through them, to study more than otherwise, and feel more emotionally attached than in a regular guidebook or sign (Jung et al., 2016). AR provides tourists with information right at their location making them know the purpose of the landmark.

Despite such advantages, most AR tourism applications do not go beyond exhibiting static images. Visual cues to initiate the display of content are used by most of them, such as QR codes or special pictures. Although convenient, markers do not allow people to discover their surroundings due to the need to find the marker and walk around rather than walk freely. No map AR allows an individual to move more realistically, yet it is ineffective in many cases on phones due to the absence of power resources, sometimes incompatible with other lighting, and is shaky outdoors (Rauschnabel et al., 2022).

The same information is also posted on everybody in most of the AR tourism apps. It is typically predetermined and has nothing to do with who you are and what you have a liking to. These applications are not much involved in smart content creation or talkback, thus they do not assist in deep learning. Such a failure to adapt creates a large hole which is being occupied by adding AI which identifies the user and negotiates with him.

0.5 Computer Vision for Landmark Recognition

Computer Vision (CV) is instrumental in enabling computer-based systems to interpret the environment. In landmark recognition early CV also utilized hand-crafted features such as SIFT and ORB, matching features across pictures (Lowe, 2004; Rublee et al., 2011). These techniques had been successful in laboratory conditions but failed in a disorganized environment, obscured areas of objects, and large sets of images.

The concept of deep learning has enhanced recognition of landmarks. In most different settings, Convolutional Neural Networks (CNNs) are able to extract complex visual features and identify locations in an accurate manner. Nevertheless, most of the powerful CNNs require much computer resources and cannot execute readily on the devices that tourists typically carry.

This was addressed by the creation of light weight models such as MobileNet. They provide an acceptable balance between speed and accuracy (Howard et al., 2017). These are the models that suit the phones and other miniature gadgets, and therefore can be used in real-time landmark recognition in the tourist field. Nevertheless, in heritage tourism sites, most studies have been conducted on large benchmark datasets rather than

using it practically.

The other issue is that CV landmark recognisers are often disjointed software. Another way is that in most applications, recogniser output is not introduced to the AR display or a speech recognition assistant, and appears separately. The recogniser itself does not facilitate the learning or interaction of people when it is not interconnected in a smooth way. This is a void that demonstrates why we require complete pipelines that include recognition, visual display and interaction (Zhu et al., 2020)

0.6 NLP-Based Assistants in Tourism

NLP technology is finding its way into the tourism industry e.g. to provide customer care services, book rooms and plan. NLP-based chatbots are able to provide answers to questions automatically and streamline and simplify the process of service (Xiang et al., 2021). However, they are barely utilized on the ground in heritage sites.

The majority of tourism chatbots do not correspond to what the user is viewing at the moment and respond in general. An explanation should be what the visitor can see in a heritage location. The recent larger language models are capable of talking fluently and providing good answers, and finally, bridging them with live vision and AR input is uncommon.

In addition, a large number of NLP systems are dependent on cloud APIs that require a stable internet connection. Connection may be poor in such destinations as Kathmandu due to network or regulation of site protection. This complicates the process of integrating functional speaking assistants in the culture sites. So we require the hybrid solutions which would integrate online chatting and the offline fallback solutions which would provide individuals with the opportunity to resort to the assistant when the network would become weak.

0.7 Tourism Recommendation Systems

Recommendation systems are one of the most popular in the tourism sector though they offer individuals personal recommendations of what there is to see, go, or do. The collaborative and content based filtering are the two primary forms of recommendation to use (Ricci et al., 2015). Collaborative filtering involves using previous actions of other users, ratings and habits to make suggestions. It can be effective on large digital sites where a great number of users are present, but it is not as effective on heritage tourism that has minimal or no user data.

Content-based filtering performs the recommendations based on facts about an item as its time period, style, cultural meaning, or location. This method is more consistent

with cultural tourism since it does not require a high amount of user data and can make use of knowledge on the part of the experts.

Most of the current tourism recommenders do not involve real time context. They are often operated within a mobile application or a webpage and do not relate to AR or CV that understands what the visitor is looking at. In this way, the recommendations may become incongruent. The combination of content-based recommendation and landmark recognisers and AR overlay may provide context-sensitive, personalized advice varying with the visitor position across a heritage site.

0.8 Critical Analysis and Gap Identification

There are three large gaps identified in the literature which render tech-based tourism tools weaker. The lacks indicate technical and conceptual issues in the application of new technology to cultural heritage.

The largest gap is that we are lacking a single application that would combine AR, CV, and NLP. Placing historical information on places has been a common application of AR, however, these overlays are usually fixed and you need to activate them manually. CV based systems that detect landmarks are typically only individual components that only verify that something is correct, they do not display or allow one to engage with it. NLP helpers tend to only be chat windows that do not have any connection to the surrounding environment. Due to this division, individuals are compelled to switch in-between various tools or windows, as a result of which this experience is fragmented, immersive, and not very practical whenever they visit actual heritage locations.

The second gap is that chatbots in the tourist industry provide generalized responses that are not related to what the tourist can see. Even though there are numerous services that rely on chatbots, the majority of such robots provide programmed answers that do not depend on the environment that any visitor is in. These bots have no idea what the user is looking at or his whereabouts, therefore they are unable to update their responses when an additional monument shows. This prevents meaningful, question-based trips amongst tourists, such as a follow-up question concerning one of the buildings or features. Due to this reason, AI chat helpers are unable to teach and inform tourists as much as they do with cultural tours.

The third gap is that the AR heritage applications tend to present the same information to all the users. These applications do not take into account the interests, beforehand knowledge, or the way of exploration of a visitor. The problem with a one size fits all strategy is that it cripples interaction and provides a shallow understanding of the culture. AR applications cannot actually provide deeper learning or longer engagement of visitors, as there are no means to personalize the material.

Identifying solutions to such issues requires a holistic resolution which will incorporate

recognition, provision of information, interactivity, and customisability into one smart tourism guide.

0.9 Project Justification

The tech in this project is comprised of Computer Vision (CV), Augmented Reality (AR) and Natural Language Processing (NLP) all in a single mobile application, since present-day tourism technologies are divided. Combined with everything, the application is able to identify landmarks, display AR content, and interact through one flow with users. This facilitates the experience and makes it easy to tourists.

We are automating heritage site identification features of CV such that the user does not need to do anything special. CV allows user to simply point the phone at a structure rather than the marker based technology which requires the use of QR codes. Our choice of small deep learning models is to enable it to run in a short period of time on the phone. The number of big heritage sites that can be identified within the Kathmandu Valley ethnic area in the app will be limited to ten so that the data set will be simple, more accurate, and compatible with varying weather conditions.

AR overlays are not marked to ensure that they do not require special tags. They present information directly in the sight of the user in a more natural feeling. To ensure the phone does not run out of battery or slow down, the AR is uncomplicated, which are text labels, informative cards, and simplified icons. We also avoid 3-D models which consumed too much computing power, data and which would have been difficult to complete in time.

The chat helper of NLP is included to ensure that the people are able to ask questions and receive more than just simple facts. Online API would provide fast answers to queries and consumed a lot of computing power which is why a fully offline chat model would require a lot of computing power. We are also adding a few offline responses to our likely questions so that the application will not black out when the internet is not strong.

Lastly, our recommendation system will also be a content based system since not much information about visitors to heritage tourism in Kathmandu is available. This system relies on facts concerning landmarks such as when the landmark was constructed, the type of culture and their proximity to the user to recommend places the user may be interested in. This is quite appropriate with culture travel, requires no excessive user information, and assists the visitors in a smart way to explore.

Each of these options attempts to compromise between emerging concepts, feasible constraints, and concrete scholarly research, therefore the application could be built on a final-year project, and will be very suitable to tourism in Kathmandu.

0.10 Project Aims, Objectives, and Scope

0.11 Project Aim

The proposed project is the design and construction of a phone-based smart virtual tour guide app. It is based on computer vision, markerless augmented reality, natural language processing and contents based recommendations. The application will provide context-sensitive cultural navigation to individual foreign tourists in the Kathmandu Valley heritage sites.

0.12 Project Objectives

The project objectives are defined according to the SMART criteria as follows:

1. To Create a computer vision that is capable of identifying ten or more UNESCO or international heritage sites in the Kathmandu Valley with a smartphone camera. By the completion of the development phase, the system should have a minimum 80 percent accuracy.
2. To Design an Augmented Reality teamwork, with no markers, displaying the context menuing on the landmark which may be historic facts or comments within three seconds after successful landmark identification.
3. To integrate an API-based NLP conversational assistant that provides real-time, context-aware responses to tourist queries in English, achieving at least 70% response accuracy based on manual evaluation using predefined test questions.
4. To design and implement a content-based recommendation system that suggests relevant nearby heritage sites based on user interactions and inferred interests during application usage.
5. To develop a fully functional Android mobile application that cohesively integrates landmark recognition, AR overlays, chatbot interaction, and recommendations within the project timeline.

0.13 Project Scope

This project is geographically constrained to UNESCO and major cultural heritage sites located within the Kathmandu Valley. The application is designed for Android smartphones and targets independent foreign tourists, while remaining applicable to domestic visitors.

Included Features

- **Mobile-based landmark recognition using TensorFlow Lite:** Included to enable real-time, on-device identification of heritage landmarks using smartphone cameras while maintaining low computational overhead.
- **Markerless AR overlays for cultural and historical information:** Included to provide immersive, contextual information directly over physical landmarks without requiring artificial markers or QR codes.
- **NLP-based conversational assistant (English only):** Included to allow users to ask natural language questions and receive real-time, context-aware responses, enhancing interactivity and engagement.
- **Content-based recommendation system:** Included so that tourists might discover landmarks that suit their preferences based on their previous behavior without using giant user databases.
- **Android mobile deployment:** Included due to the widespread availability of ARCore-supported Android devices and better compatibility with selected development tools.

Excluded Features

- **3D reconstruction and photogrammetry:** Extra since it requires a high computing power, substantial amount of data and at the same time it is not very practical in providing instant cultural assistance.
- **On-device large language models:** Excluded because current mobile devices lack sufficient processing capability to run large-scale NLP models efficiently without impacting performance and battery life.
- **Global landmark datasets beyond Kathmandu Valley:** World archives outside Kathmandu Valley: Omitted to ensure the project maintains focus, make recognition more precise and be able to complete on time.

0.14 Project Design

0.15 Design and Methods

The project employs an iterative project development which passes through the data collection, model building processes, system integration and testing. Each stage test ensures each of the components is functioning and is performing as required.

0.15.1 Data Collection

Field visits to the Kathmandu Valley will be used to capture image data of heritage landmarks to capture the remainder of the viewpoints and lighting conditions of the buildings with the help of smartphone cameras. The data will be enhanced with open-source pictures where necessary. The sources of metadata, including historical descriptions and cultural significance, will be found on reputable sites, including Wikipedia and the Department of Archaeology, Nepal.

0.15.2 Computer Vision Model Development

Convolutional Neural Network based on MobileNetV2 will be used to train with the help of the TensorFlow and ten different heritage landmarks will be classified. The ready model will be translated to the TensorFlow Lite format so that it can be used with inference on the device. Information distortion like rotation and brightness alteration will be used to advance the strength of the model

0.15.3 Augmented Reality Implementation

The ARCore will be used to implement the Markerless AR overlays, through a Flutter AR plug-in. Upon detecting a landmark, cards of contextual information and annotations will be displayed on the screen of the device and aligned with the real world. The overlay complexity will remain minimal and warrant real-time performance and less battery consumption.

0.15.4 Chatbot Integration

A conversational assistant using an API-based Large Language Model will be included to respond in real-time to the situation. The chatbot will be provided with metadata of the detected landmark name and the corresponding metadata. Limited network connectivity will be supported through a local fallback based on a FAQ.

0.15.5 Recommendation System

An establishment of an appraisal system of content-based recommendation will be constructed based on already defined land marks features like historical classification, values, and the physical location. The reactions of the users towards the application will be used to draw conclusions about the likes and propose appropriate future destinations.

0.15.6 Usability and System Testing

System evaluation will involve landmark recognition significance with unseen images, AR response time, chatbot recognition with 30 predefined test questions as well as brief user testing to determine relevance of the recommendation and general usability.

0.16 Justification of Methods

MobileNetV2 is selected as the model is not very large and also as it compromises between the amount and the accuracy of classification combined with computational efficiency, therefore, it can be used in real-time by mobile applications. TensorFlow Lite enables Android to perform inference quickly and possesses the advantage of a small performance footprint.

Flutter is employed on the creation of apps due to its fast UI construction and solidness of the cataclysmic multiplicity of a shipment. ARCore is also better than options such as Vuforia since it has native Android functionality, powerful markerless tracking, and reduced licensing and integration.

A NLP conversational assistant is chosen based on API and offers better quality of natural language responses without the extreme burden of computation that on-device language models impose. The reason behind choosing content-based filtering over collaborative one is that no big user set is required and it is appropriate with the structured cultural heritage attributes.

In general, the selected methods will ensure the possibility of technical scalability and adaptation to the goals of the project regardless of time constraints and available resources.

The project can be done owing to the availability of readily available tools, including Flutter, Firebase, TFLite, and ARCore, and collecting data in Kathmandu being manageable. It is possible to train a model in Google Colab or university GPUs, which aligns with the abilities and academic capabilities of the student.

0.17 Project Timeline

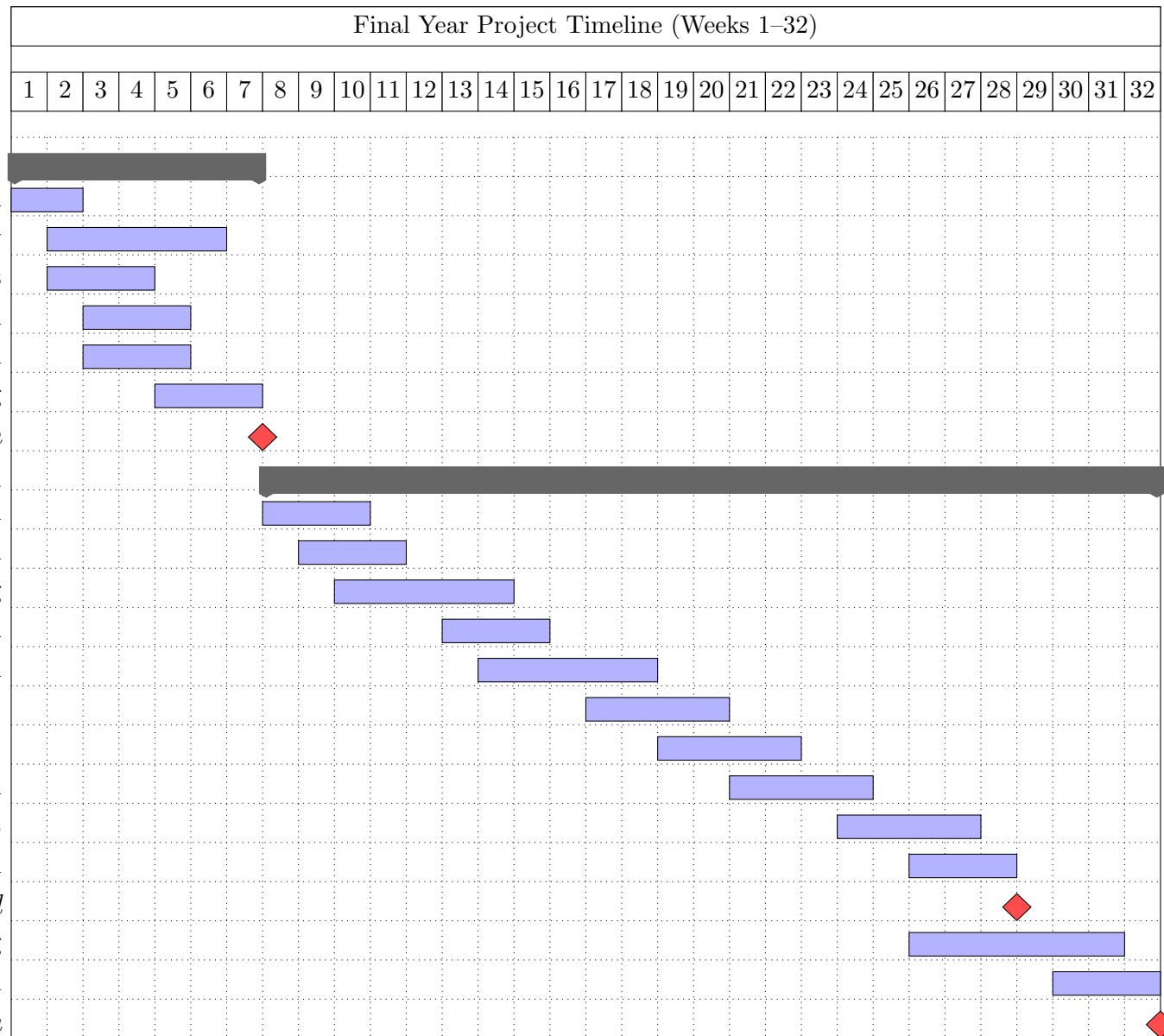


Figure 1: Gantt chart showing the project timeline and key milestones

0.18 Feasibility, Risks, and Ethical Aspects

0.18.1 Feasibility

The project is feasible due to the availability of accessible tools (Flutter, Firebase, TFLite, ARCore) and manageable data collection requirements within Kathmandu. Model training can be completed using Google Colab or university GPUs. The technical scope aligns with the student’s skill set and academic resources.

0.18.2 Risk Analysis and Mitigation

Risk	Likelihood	Impact	Mitigation
Insufficient training data	Medium	High	Collect images from multiple angles and use augmentation.
ARCore compatibility issues	Medium	Medium	Test on real devices and use stable packages.
API limitations for chatbot	Medium	Medium	Implement an offline FAQ fallback system.
Time constraints	High	High	Follow a strict milestone schedule and prioritize core features.
Ethical concerns	Low	Medium	Avoid collecting personal data and obtain consent for testing.

0.18.3 Ethical Considerations

The project will ensure ethical compliance by protecting user privacy, securing API communications, and obtaining informed consent for any user testing. No personal or sensitive data will be collected. All images and cultural information will be sourced ethically from publicly available datasets.

0.19 References

- Azuma, R. (1997) A survey of augmented reality. *Presence*, 6(4), pp. 355–385.
- Bakas, C. et al. (2019) Mobile augmented reality in cultural heritage. *Journal of Cultural Heritage*, 38, pp. 1–9.
- Chung, N. et al. (2021) Augmented reality and AI in tourism. *Tourism Management*, 83, 104242.
- Howard, A. et al. (2017) MobileNets: Efficient CNNs for mobile vision applications. *arXiv preprint arXiv:1704.04861*.

- Jung, T. et al. (2016) The influence of augmented reality on tourism experience. *Journal of Travel Research*, 55(8), pp. 1024–1040.
- Lowe, D. (2004) Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2), pp. 91–110.
- OECD (2020) *Tourism Trends and Policies*. Paris: OECD Publishing.
- Ricci, F., Rokach, L. and Shapira, B. (2015) *Recommender Systems Handbook*. Springer.
- Xiang, Z. et al. (2021) Chatbots in tourism. *Tourism Management Perspectives*, 37, 100780.